MODEL TRENING TAQQOSLASHI

KETMA-KETDAN PARALLEL REDUCE'GACHA

Fraud aniqlash modelida tezlik, aniqlik va xatolar muvozanati

PaySim • Random Forest, 100 daraxt • 16 CPU worker

ASOSIY NATIJA

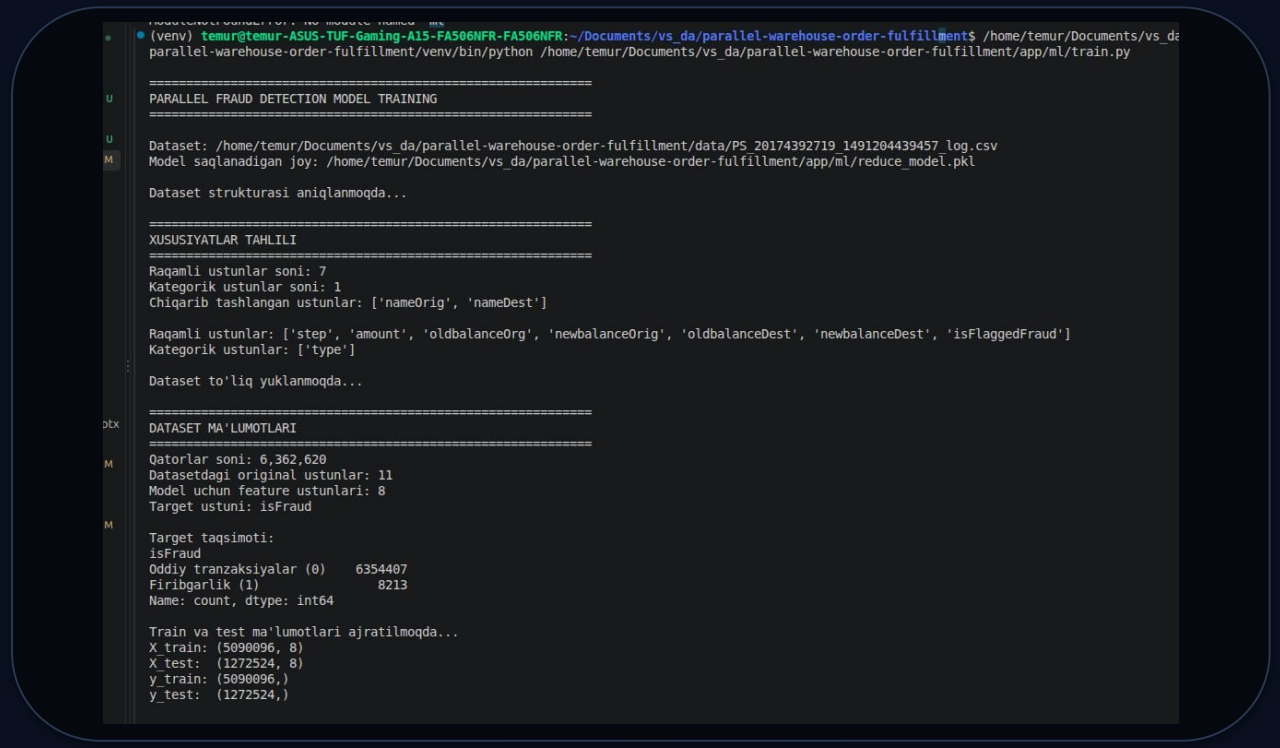
15.6×

tezroq o‘qitish
Parallel → Parallel Reduce

Ketma-ket natija logda yo‘q — taxminiy raqam kiritilmadi.

Tajriba ma’lumotlari: PaySim

Bir xil holdout test to‘plami natijalarni solishtirishga imkon beradi.



JAMI TRANZAKSIYA

6.36 mln

CSV: 6,362,620 qator

FRAUD HOLATI

8,213

Ulushi: 0.129%

MODEL FEATURE

8

7 raqamli + 1 kategorik

TRAIN / TEST

80 / 20

Stratified ajratish

Test: 1,272,524 qator • fraud: 1,643

ID ustunlari chiqarilgan: nameOrig, nameDest

Muhim: sinflar juda nomutanosib. Shu bois accuracy’ni yolg‘iz baholash yetarli emas; fraud sinfiga precision, recall va F1 ham qaraladi.

USULLAR

Uch xil o‘qitish yondashuvi

Model sifati uchun asosiy mezon — bir xil feature, split va test to‘plamida taqqoslash.

01 KETMA-KET

1 worker • n_jobs=1

Daraxtlar bitta CPU oqimida quriladi. Loyiha tarixidagi kodda bu rejim bor, ammo yuborilgan natijada uning vaqti ham, metrikalari ham yozilmagan.

02 PARALLEL

16 worker • 100 daraxt

Oddiy Random Forest: daraxtlarni CPU workerlar bir vaqtda quradi. Skrinshotda ketma-ket rejim o‘chirib qo‘yilgan, parallel natija o‘lchangan.

03 PARALLEL REDUCE

16 worker • 100 daraxt

Train qismi stratified bo‘laklarga ajratiladi; 16 model 6–7 daraxtdan o‘qitilib, fraud ehtimollari average’lanadi.

Izoh: Reduce — faqat tezroq bajarish emas; har bir kichik model train ma’lumotining bir bo‘lagini ko‘radi, shuning uchun modelning o‘rganish usuli ham o‘zgaradi.

VAQT TAHLILI

O‘qitish tezligi: Reduce 15.6× tezroq

Skrinshotlardagi fit bosqichi vaqtlarini solishtirish — eng to‘g‘ri speed benchmark.



FIT BOSQICHI

15.6×

189.95 / 12.18 sekund

UMUMIY RUN

5.16×

197.87 / 38.32 sekund

REDUCE FIT VAQTI

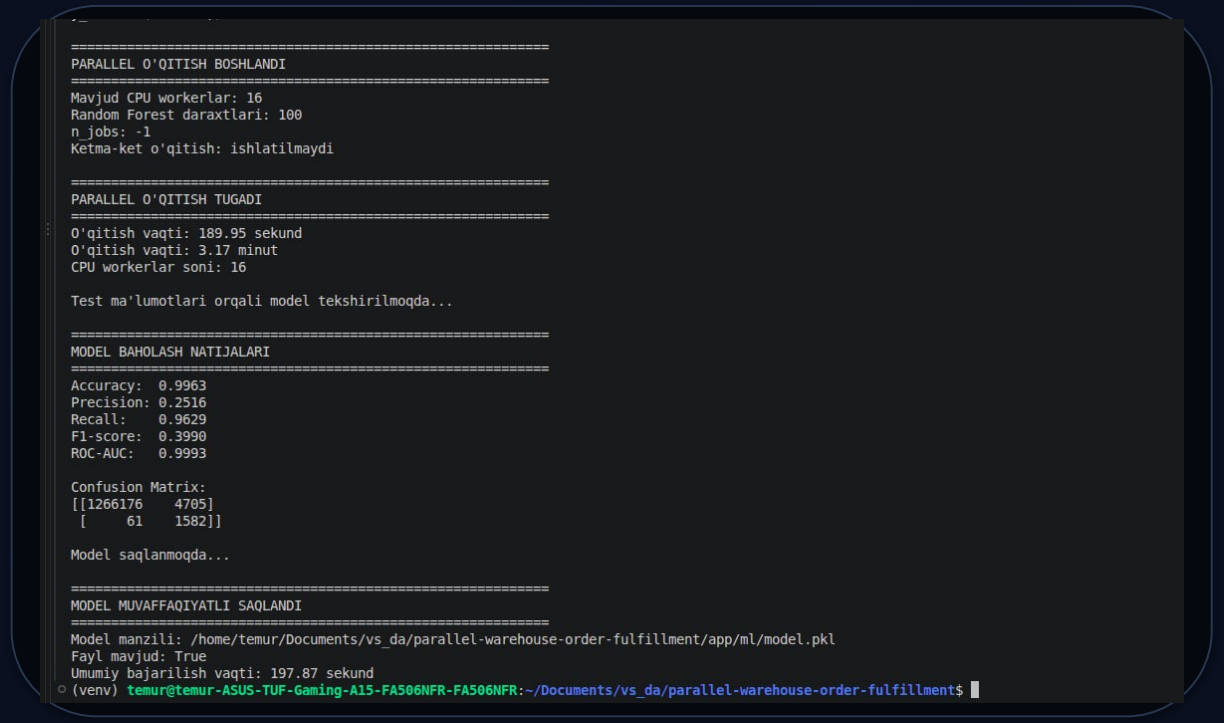
−93.6%

parallel fit vaqtiga nisbatan

Umumiy run vaqtiga CSV yuklash, test va saqlash ham kiradi; bu vaqt fit vaqtidek aniq bir xil chegaralarda o‘lchanmagan bo‘lishi mumkin. Ketma-ket vaqt mavjud emas.

Oddiy parallel o‘qitish natijasi

16 CPU worker • Random Forest: 100 daraxt • test to‘plami: 1,272,524 tranzaksiya



```
============================================================
PARALLEL O'QITISH BOSHLANDI
============================================================
Mavjud CPU workerlar: 16
Random Forest daraxtlari: 100
n_jobs: -1
Ketma-ket o'qitish: ishlatilmaydi

============================================================
PARALLEL O'QITISH TUGADI
============================================================
O'qitish vaqti: 189.95 sekund
O'qitish vaqti: 3.17 minut
CPU workerlar soni: 16

Test ma'lumotlari orqali model tekshirilmoqda...

============================================================
MODEL BAHOLASH NATIJALARI
============================================================
Accuracy:  0.9963
Precision: 0.2516
Recall:    0.9629
F1-score:  0.3990
ROC-AUC:   0.9993

Confusion Matrix:
[[1266176    4705]
 [     61    1582]]

Model saqlanmoqda...

============================================================
MODEL MUVAFFAQIYATLI SAQLANDI
============================================================
Model manzili: /home/temur/Documents/vs_da/parallel-warehouse-order-fulfillment/app/ml/model.pkl
Fayl mavjud: True
Umumiy bajarilish vaqti: 197.87 sekund
(venv) temur@temur-ASUS-TUF-Gaming-A15-FA506NFR-FA506NFR:~/Documents/vs_da/parallel-warehouse-order-fulfillment$
```

ACCURACY

99.63%

PRECISION

25.16%

RECALL

96.29%

F1-SCORE

39.90%

ROC-AUC

99.93%

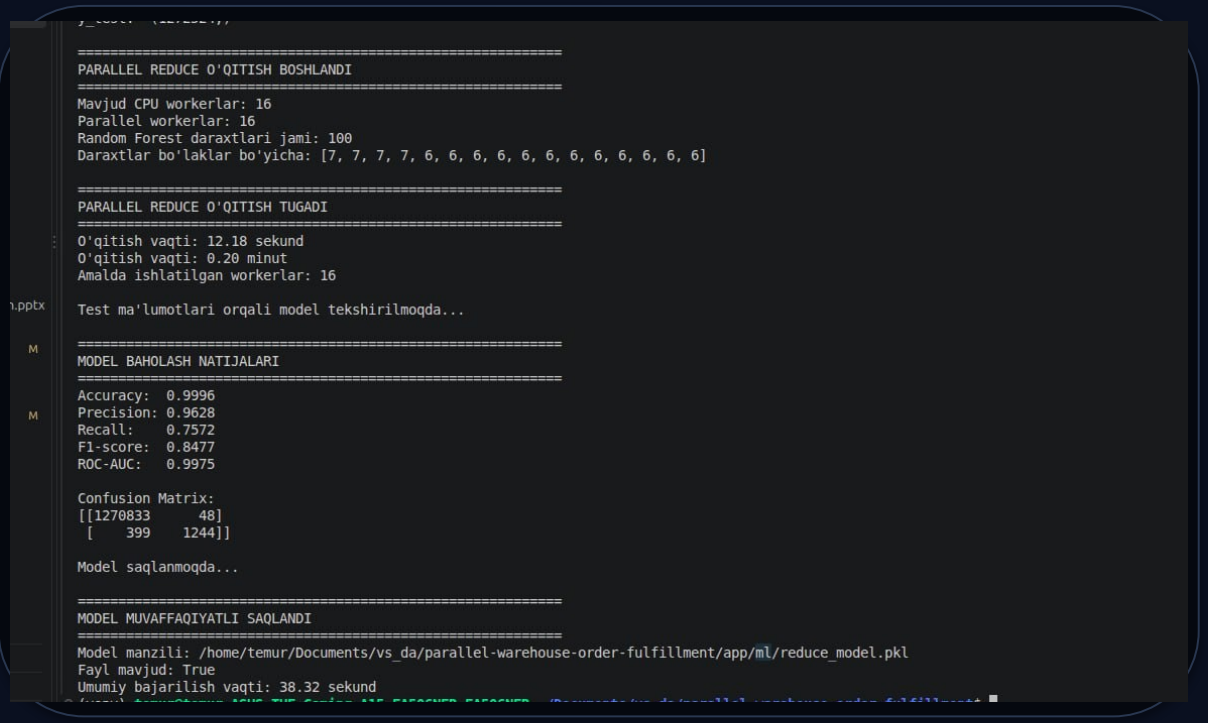
CONFUSION MATRIX

FP 4,705 FN 61

Talqin: fraudlarning 96.29% topilgan, lekin fraud deb belgilangan tranzaksiyalarning atigi 25.16% i haqiqiy — false positive ko‘p.

Parallel Reduce natijasi

16 parallel bo‘lak • jami 100 daraxt • bir xil test hajmi



```
============================================================
PARALLEL REDUCE O'QITISH BOSHLANDI
============================================================
Mavjud CPU workerlar: 16
Parallel workerlar: 16
Random Forest daraxtlari jami: 100
Daraxtlar bo'laklar bo'yicha: [7, 7, 7, 7, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6, 6]

============================================================
PARALLEL REDUCE O'QITISH TUGADI
============================================================
O'qitish vaqti: 12.18 sekund
O'qitish vaqti: 0.20 minut
Amalda ishlatilgan workerlar: 16

Test ma'lumotlari orqali model tekshirilmoqda...

============================================================
MODEL BAHOLASH NATIJALARI
============================================================
Accuracy:  0.9996
Precision: 0.9628
Recall:    0.7572
F1-score:  0.8477
ROC-AUC:   0.9975

Confusion Matrix:
[[1270833      48]
 [    399    1244]]

Model saqlanmoqda...

============================================================
MODEL MUVAFFAQIYATLI SAQLANDI
============================================================
Model manzili: /home/temur/Documents/vs_da/parallel-warehouse-order-fulfillment/app/ml/reduce_model.pkl
Fayl mavjud: True
Umumiy bajarilish vaqti: 38.32 sekund
```

ACCURACY

99.96%

PRECISION

96.28%

RECALL

75.72%

F1-SCORE

84.77%

ROC-AUC

99.75%

CONFUSION MATRIX

FP 48 FN 399

Talqin: false positive keskin kamaygan va F1 yaxshilangan; evaziga fraudlarni o‘tkazib yuborish 61 tadan 399 taga ko‘paygan.

TO‘G‘RIDAN-TO‘G‘RI TAQQOSLASH

Metrikalar: qaysi ko‘rsatkich o‘zgardi?

Ikkala confusion matrix ham aynan 1,270,881 normal va 1,643 fraud test holatiga tegishli.

METRIKA	PARALLEL	REDUCE	O‘ZGARISH
Accuracy	99.63%	**99.96%**	+0.33 foiz punkt
Precision	25.16%	**96.28%**	+71.12 foiz punkt
Recall	96.29%	**75.72%**	−20.57 foiz punkt
F1-score	39.90%	**84.77%**	+44.87 foiz punkt
ROC-AUC	99.93%	**99.75%**	−0.18 foiz punkt

Xatolar almashinuvi: FP 4,705 → 48 (−99.0%); FN 61 → 399. Bitta umumiy “aniqlik” soni biznes uchun qaysi model afzal ekanini o‘zi hal qilmaydi.

Natijaning amaliy ma’nosi: xatolar narxi

Fraudni o‘tkazib yuborish va qonuniy tranzaksiyani to‘xtatishning xarajati bir xil emas.

FALSE POSITIVE

4,705 → 48

Legitim operatsiyani shubhali deb to‘xtatish

FALSE NEGATIVE

61 → 399

Fraud tranzaksiyani oddiy deb o‘tkazib yuborish

REDUCE AFZALLIGI

Yuqori precision va F1; kamroq noto‘g‘ri ogohlantirish.

REDUCE XAVFI

Recall pastroq: fraudlarning chorakka yaqini topilmay qoladi.

Accuracy ≈99.9% bo‘lishi ko‘p normal tranzaksiyalar ulushi bilan izohlanadi. Threshold, real fraud qiymati va mijozga ta’sir alohida baholanmasdan production tanlovi qilinmasin.

YAKUNIY XULOSA

Xulosa va keyingi to‘g‘ri qadam

Tezlik yutug‘i katta, ammo model tanlovi fraudni topish va xato ogohlantirish narxiga bog‘liq.

ENG TEZ O‘LCHANGAN

Reduce

12.18 s fit • 15.6× speedup

ENG YUQORI RECALL

Parallel

96.29% • FN = 61

ENG YUQORI F1

Reduce

84.77% • FP = 48

KETMA-KET BASELINE — O‘LCHOV YO‘Q

Loyiha tarixidagi train.py da n_jobs=1 rejimi bor, biroq hozirgi skrinshotlar bu running vaqtini yoki metrikalarini bermaydi. Ketma-ket natijani aniq taqqoslash uchun qayta o‘lchash kerak.

TAQQOSLASHNI YOPISH UCHUN

Bir xil split/seed/feature va 100 daraxt bilan n_jobs=1 ni o‘lchang; fit vaqtini alohida, barcha metrikalarni bitta testda yozib oling.

Tanlov: recall muhim bo‘lsa — parallel; precision/F1 va tezlik muhim bo‘lsa — Reduce’ni threshold sozlab tekshirish. Haqiqiy tranzaksiyalar bilan mustaqil validatsiya zarur.